$$y = x\frac{dy}{dx} - ln\frac{dy}{dx}$$

$\frac{dy}{dx} = p$

$y = xp - \ln p$

$\frac{dy}{dx} = p + x\frac{dp}{dx} - \frac{1}{p}\frac{dp}{dx}$

$\cancel{p} = \cancel{p} + \frac{x\,dp}{dx} - \frac{1}{p}\frac{dp}{dx}$

$0 = \left(x - \frac{1}{p}\right)\frac{dp}{dx}$

$p = c$

$y = xc - \ln c$

$x - \frac{1}{p} = 0$

$p = \frac{1}{x}$

$y = 1 - \ln\frac{1}{x}$

$y = 1 + \ln x$

1) $\left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$

$\frac{dy}{dx} = p$

$y = xp - p^2$

$\frac{dy}{dx} = x\frac{dp}{dx} + p - 2p\frac{dp}{dx}$

$\cancel{p} = x\frac{dp}{dx} + \cancel{p} - 2p\frac{dp}{dx}$

$0 = x\frac{dp}{dx} - 2p\frac{dp}{dx}$

$0 = (x - 2p)\frac{dp}{dx}$

$p = 0 + c$

$y = xc - c^2$

$y = \frac{x^2}{2} - \frac{x^2}{4}$

$y = \frac{x^2}{4}$

$x - 2p = 0$

$x = 2p$

$\frac{x}{2} = p$

$x^2 = 4y$

2) $y - x\frac{dy}{dx} = 3\left(\frac{dy}{dx}\right)^2$

$\frac{dy}{dx} = p$

$y - xp = 3p^2$

$\frac{dy}{dx} = x\frac{dp}{dx} + p + 6p\frac{dp}{dx}$

$\cancel{p} = x\frac{dp}{dx} + \cancel{p} + 6p\frac{dp}{dx}$

$0 = x\frac{dp}{dx} + 6p\frac{dp}{dx}$

$0 = \frac{dp}{dx}(x + 6p)$

$0 = x + 6p$

$p = c$

$-\frac{x}{6} = p$

$y - xc = 3c^2$

$y - x\left(-\frac{x}{6}\right) = 3\left(-\frac{x}{6}\right)^2$

$36/3 = 12$

$y = -\frac{3x^2}{36} - \frac{x^2}{6}$

$-\frac{1}{12} - \frac{2}{12} = -\frac{3}{12} = -\frac{1}{4}$

$y + \frac{x^2}{6} = -\frac{3x^2}{36}$

$y = -\frac{x^2}{4}$